AppWorld Topic Distributions and Scenario Samples

1 Train and test topic distributions

An AppWorld scenario is a family of three task variants sharing an ID prefix. The full train, normal-test, and challenge-test splits contain 30, 56, and 139 scenarios, respectively. We describe task content using ten instruction-derived topics: Amazon, Gmail, Todoist, Splitwise, Spotify, Venmo, Simple Note, Phone, File system, and Other. App topics require an explicit app mention; Phone and File system also use corresponding instruction references. An instruction without any of these tags is assigned Other. This is a coarse description of task content: for example, an email task that does not explicitly mention Gmail may be tagged Other.

Each scenario has total weight one. Its three variants each contribute one third of that weight, divided equally among all topics tagged in that variant. Let $T_{s,v}$ be the set of tags in variant v of scenario s. For a split G, the topic probability is

$$p_{G,k} = \frac{1}{|G|} \sum_{s \in G} \frac{1}{3} \sum_{v=1}^{3} \frac{\mathbf{1}\{k \in T_{s,v}\}}{|T_{s,v}|}, \qquad \sum_{k=1}^{10} p_{G,k} = 1. \tag{1}$$

The indicator $\mathbf{1}\{k \in T_{s,v}\}$ is one if topic k is tagged in that variant and zero otherwise. The denominator $|T_{s,v}|$ is the number of tags in the variant; it is at least one because an untagged instruction receives Other. For example, if $T_{s,v} = \{\text{Spotify}\}$, the fraction is one for Spotify and zero for every other topic. If $T_{s,v} = \{\text{Spotify}, \text{Phone}\}$, it is $1/2$ for each of those two topics and zero for the rest. That variant therefore contributes $(1/3)(1/2) = 1/6$ of its scenario's weight to each topic, before averaging over the $|G|$ scenarios.

Table 1: Topic probabilities in the full splits, quick2h samples (five scenarios per split), and intermediate samples (10 train, 19 normal, and 46 challenge scenarios). Each column sums to 100% before rounding.

Topic	Train			Test normal			Test challenge		
	Full	Quick2h	Intermediate	Full	Quick2h	Intermediate	Full	Quick2h	Intermediate
Amazon	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	43.65%	43.33%	44.20%
Gmail	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	11.87%	10.00%	11.96%
Todoist	0.00%	0.00%	0.00%	3.57%	0.00%	2.63%	0.72%	0.00%	0.00%
Splitwise	0.00%	0.00%	0.00%	3.57%	10.00%	5.26%	1.08%	0.00%	1.09%
Spotify	50.00%	50.00%	50.00%	21.43%	20.00%	21.05%	6.35%	6.67%	5.80%
Venmo	20.00%	20.00%	20.00%	34.52%	30.00%	34.21%	7.19%	10.00%	7.61%
Simple Note	8.33%	10.00%	5.00%	18.75%	20.00%	18.42%	3.24%	0.00%	3.26%
Phone	8.33%	0.00%	10.00%	9.23%	10.00%	10.53%	6.83%	0.00%	6.52%
File system	13.33%	20.00%	15.00%	7.14%	10.00%	7.89%	10.43%	10.00%	10.87%
Other	0.00%	0.00%	0.00%	1.79%	0.00%	0.00%	8.63%	20.00%	8.70%

In the full splits, normal test has relatively more Venmo and Simple Note content and introduces Todoist and Splitwise topics. Challenge test has substantial Amazon and Gmail content, both absent from the training instructions under these tags.

2 Total variation distance

For two topic distributions $P = (p_1, \ldots, p_{10})$ and $Q = (q_1, \ldots, q_{10})$, total variation (TV) distance is

$$\mathrm{TV}(P, Q) = \frac{1}{2} \sum_{k=1}^{10} |p_k - q_k|. \tag{2}$$

The factor 1/2 keeps the result between zero and one. A value of 0 means the two distributions assign exactly the same probability to every *measured topic*; their individual tasks need not be identical. A value of 1 means the distributions have disjoint topic support: no measured topic has positive probability in both. Equivalently, TV is the minimum fraction of topic probability mass that must be reassigned to make the distributions match.

Applying Eq. (2) to the full-split proportions in Table 1 gives

$$\mathrm{TV}(P_{\text{train}}, P_{\text{normal}}) = \tfrac{1}{2}(0.695238) = \mathbf{0.347619}, \tag{3}$$

$$\mathrm{TV}(P_{\text{train}}, P_{\text{challenge}}) = \tfrac{1}{2}(1.318945) = \mathbf{0.659472}. \tag{4}$$

These calculations use the unrounded probabilities; the displayed topic percentages are rounded to two decimal places. The challenge split is therefore much farther from train on this topic representation than normal test is.

3 Two-hour sampling methods

Quick2h sample. From each source split, select five complete scenarios and retain all three variants of each scenario: 15 scenarios and 45 tasks overall. Selection balances the split's instruction-derived topic mix, task difficulty, and number of apps. The normal and challenge samples also span different levels of instruction similarity to training scenarios and include test content absent from train. At the pilot estimate of 2.2 minutes per task, 45 tasks take about 99 minutes.

Conditioned sample. Select five complete scenarios per split under an additional memory-pressure eligibility requirement. Each family is scored from its task variants and evaluator conditions on memory structure (M), retrieval (I), adaptation (A), evidence sampling (P), and memory update (U), with scores in $\{0, 1, 2\}$. A family is eligible only if all five scores are at least one. Among reviewed eligible families, selection aims to retain each original split's topic mix, difficulty, app count, and test novelty mix. Baseline success or failure is not used as a screening criterion. The resulting 15 scenarios again contain 45 tasks. Conditioning and the five-scenario budget can change the observed topic proportions, so neither sample is an unbiased estimate of full-split performance.

3.1 Reusable total-variation-preserving selection criteria

For a new sample, fix the source splits and topic rule, then select complete three-task families within the budget. The ordinary pool is each full split; the conditioned pool contains families passing all five tests in Table 2. In both cases, compare each sample against its *full* source split for topic, difficulty, app-count, and test-novelty-bin TV. Also compare each sampled train-to-test topic TV with its full-split counterpart: for $t \in \{\text{normal}, \text{challenge}\}$, $\delta_t = |\,\mathrm{TV}(P_{S_{\text{train}}}, P_{S_t}) - \mathrm{TV}(P_{F_{\text{train}}}, P_{F_t})|$, where F denotes a full split and S its sample. Prefer small δ_t, retain the challenge-greater-than-normal ordering and test-only app content where feasible, and report any mismatch after conditioning. Detailed criteria and audit fields are in `samples/memory_pressure_screening.md`.

3.2 Memory-pressure scoring tests

Table 2 converts the reusable conditioned-sample screening rubric into score definitions. M, I, A, P, and U denote memory structure, retrieval, adaptation, evidence sampling, and memory update. Inspect all three variants and intrinsic evaluator conditions together. Each dimension receives exactly one score from $\{0, 1, 2\}$, supported by a concrete family observation and a one-sentence justification.

Table 2: Memory-pressure scoring tests used by the conditioned sample.

Mode	Score = 0	Score = 1	Score = 2
M	No meaningful reusable procedure, or variants are essentially identical.	Variants share a procedure but differ in a task-specific binding, entity, condition, threshold, or object set.	Score 1 holds, and transferring a variant-specific detail could cause an error; reusable invariants must be separated from scoped details.
I	Only one obvious reusable procedure is needed, with no meaningful competing memories.	Multiple relevant reusable rules are needed, or plausible but inapplicable distractor rules exist.	Multiple relevant rules must be retrieved jointly and plausible distractors exist; missing or misretrieving one changes the procedure.
A	A retrieved generic rule can be executed almost directly.	At least one non-trivial task-specific binding, filter, entity resolution, or ordered action is required.	At least two non-trivial operations must be composed, such as predicates, set operations, aggregation, dependencies, an output contract, or a postcondition check.
P	Failure feedback is binary or ambiguous and cannot provide a useful diagnosis.	Evaluator or trajectory evidence localizes a failure and supports a reusable lesson.	Feedback identifies a root cause amid plausible competing trajectory explanations; learning requires selecting the causal evidence.
U	Earlier variants offer little or no reusable information for later variants.	At least one earlier lesson should transfer to a later variant.	Both transferable and variant-specific non-transferable information exist, testing useful updates and negative transfer.

A family is a pressure-test candidate only when all five scores are at least one. Do not award a positive score from task difficulty alone or use baseline success and failure. For P, use feedback available to the intended learner; score 2 requires an actual trajectory with plausible competing explanations.

4 Summary

Table 3: Full splits, quick2h and conditioned five-scenario-per-split samples, and the one-third intermediate sample. Topic TV compares each row to its own full source split; it is zero for the full splits by definition. The two rightmost columns compare train and test topic distributions *within the same set*; each value is shown once per three-row set.

Sets	Splits	Number of Scenario	Number of Tasks	Topic TV	Train → Normal TV	Train → Challenge TV
Full	Train	30	90	0.0000	0.3476	0.6595
	Normal	56	168	0.0000		
	Challenge	139	417	0.0000		
Quick2h Sample	Train	5	15	0.0833	0.4000	0.7333
	Normal	5	15	0.1131		
	Challenge	5	15	0.1448		
Intermediate	Train	10	30	0.0333	0.3605	0.6594
	Normal	19	57	0.0374		
	Challenge	46	138	0.0159		
Conditioned Sample	Train	5	15	0.1000	0.3000	0.7000
	Normal	5	15	0.1131		
	Challenge	5	15	0.2233		

All three sampled sets preserve the ordering of the full data: challenge is farther from train than normal test. These distances concern the ten-topic representation, not the complete distribution of scenario semantics.